OG-Core × CLEWS — The Coupling Mechanisms

A reference: the intuition, then the technical form, for each channel.
Numbers are the committed M=8 Philippine golden record.

1 The coupling in brief

OG-Core and CLEWS describe different parts of the same transition. CLEWS/OSeMOSYS is a least-cost energy-system optimiser: it represents technologies, dispatch, investment, and emissions in detail, but treats energy demand as an external input and has no households, government budget, or distributional accounts. OG-Core is an overlapping-generations general-equilibrium economy: it represents households (by age s and lifetime income j), firms (M industries), and a government budget, but has no priced energy input in firm production.

The models are therefore connected by a *staged soft link*. A small set of solved CLEWS outcomes is translated into OG-Core parameters; selected OG-Core outcomes can then be passed back to CLEWS in the next iteration. The link is staged rather than simultaneous because the models solve time differently: OG-Core moves through the transition period by period, while CLEWS optimises the full planning horizon at once.

Three objects organise every mechanism:

- **Signal** — an outcome read from a *solved* CLEWS scenario, such as an electricity price (its levelized cost), required grid investment, or emissions.
- **Wedge** — the OG-Core parameter used to carry that signal: for example the consumption tax τ^c, industry productivity Z_m, public investment α_I, mortality $\rho(s,t)$, or effective labour $e(t,s,j)$.
- **Channel** — the rule that maps one signal into one wedge, plus a diagnostic check for cases where the mapping is too large or poorly defined.

The main signal is the **energy price**: the levelized cost of electricity — the all-in average cost (capital, running costs, and fuel) of a delivered unit, reconstructed from the solved CLEWS system. In OG-Core this signal does two things CLEWS cannot do on its own: it lets quantities respond to price, and it shows which households and industries bear the cost. Because OG-Core firms do not buy energy explicitly, the electricity-price channel uses one composite transmission. It applies a cost-push to industries that use electricity as an intermediate input, and a consumption wedge to households that buy energy directly. The first part captures the contractionary, wage-bearing effect; the second captures the distributional cost-of-living effect. The input side is reduced-form: the fully structural version would add energy as a priced production input.

Throughout, a hat denotes a reform/baseline ratio ($\hat{x} \equiv x^r/x^b$), and i_e / m_e are the energy *good* / *industry* (Philippines: $m_e = 2$).

2 The mechanisms

The channels split by direction. Four carry CLEWS signals into OG-Core, two are policy levers applied consistently across the linked models, and two send OG-Core outcomes back to CLEWS.

2.1 Energy price

[CLEWS → OG]

Intuition. An electricity-price change enters the economy through production and household consumption. Industries buy most electricity as an *input* (about 73% in the Philippine accounts), so a higher price raises production costs and affects output, wages, and returns. Households buy electricity *directly* (about 25%), so the same price increase raises living costs. The proportional effect is larger for poorer households because energy takes a larger share of their budgets. This channel carries both uses together.

Technical. The reform/baseline electricity-price ratio g is applied through both uses. *(i) Firm input — cost-push:* each industry j's productivity is cut by electricity's share ϕ_j of its input costs (from the SAM), $Z_j \mapsto Z_j/(1+\phi_j(g-1))$, so its output price rises by $\approx \phi_j(g-1)$ — the contractionary, wage-bearing side. *(ii) Household consumption — wedge:* a consumption-tax wedge on the energy good, scaled to electricity's value-share σ of that good,

$$1+\tau^{c,r}_{i_e} \;=\; \big(1+\sigma(g-1)\big)\big(1+\tau^{c,b}_{i_e}\big)$$

— the regressive cost-of-living side. The additional revenue can be rebated to households as transfers when the run uses revenue recycling. The electricity industry's own purchases of electricity are removed from the cost-push to avoid double-counting through the household wedge. The price g is taken from CLEWS's automatic price source: a cost-of-electricity index if available, otherwise the levelized cost of electricity reconstructed from the run's costs and generation. In the Philippine run it rises about 5% on average (up to 16% by 2030); the standalone channel is still small (-0.006% of steady-state output).

Variables. $g = \pi^r/\pi^b$, the reform/baseline electricity-price ratio; ϕ_j, electricity's share of industry j's input costs; σ, electricity's value-share of the household energy good i_e; Z_j, industry j's productivity; $\tau^c_{i_e}$, the consumption-tax wedge on the energy good (r, b = reform, baseline).

2.2 Carbon price

[policy — both models]

Intuition. The carbon price is a policy input applied once. In the energy model, it tilts investment toward cleaner technologies. In the economy, it raises the price of household energy like a tax. It is not also inferred from CLEWS's implied carbon cost, because that would count the same policy twice.

Technical. A price cp (US\$/tCO_2) becomes an add-on to the energy-good wedge in OG-Core, and an emissions-penalty path in CLEWS:

$$\Delta\tau^c_{i_e} \;=\; \frac{cp \cdot \text{deflator} \cdot \text{carbon intensity}}{p^b_{i_e}}.$$

Because OG-Core firms do not buy energy, the macro-side wedge covers only *household* energy carbon; industrial carbon is priced in CLEWS. Revenue can be recycled as transfers (α_T). Applied by itself, this channel changes steady-state output by -0.021%; that level should be read as illustrative because the conversion from CLEWS monetary units to the OG-Core numéraire is not yet calibrated.

Variables. cp, the carbon price; the deflator maps CLEWS money to the OG-Core numéraire (currently 1, uncalibrated); carbon intensity, tCO_2 per unit of the energy good; α_T, the transfer share used to recycle revenue.

2.3 Public investment [CLEWS → OG]

Intuition. Grid expansion is public investment. If it is debt-financed, it competes for savings, raises the interest rate, and can crowd out private investment. At the same time, the resulting public capital is productive, so it can raise output. The net effect depends on which force dominates.

Technical. The public grid-capex increment (as a share of GDP) is added to the public-investment share α_I on a finite ramp; it accumulates into public capital, which is productive in the firm's technology:

$$K_{g,t+1} = (1-\delta)\,K_{g,t} + I_{g,t}, \qquad I_{g,t} = \alpha_{I,t}\,Y_t.$$

This is where crowding-out enters the linked model. In the Philippine scenario the extra grid spend is ≈ 0 because the transition is mostly generation-side, so this channel has essentially no effect here; the steady-state output change rounds to +0.000%. With non-zero grid spending, the same mechanics scale accordingly.

Variables. α_I, the public-investment share of output; K_g, the public capital stock; I_g, public investment; δ, depreciation; γ_g, public capital's share in production.

2.4 Capital intensity [CLEWS → OG]

Intuition. A clean power fleet is capital-heavy: it has high up-front capex and low fuel costs. That does not mean this channel necessarily pulls more capital into the energy sector. Here the lever raises the energy sector's capital *share* while holding productivity fixed. In a competitive model, that lowers operating cost and pushes the electricity price down. Because electricity demand responds only weakly, sector revenue falls and the sector can shed capital. This channel is therefore a factor-income and energy-price lever, not an investment-pull lever.

Technical. The reform/baseline ratio of the power fleet's capital-cost share scales the energy industry's Cobb–Douglas capital share γ_m (also its capital exponent; Philippines: +12.2%). Under Cobb–Douglas technology ($\varepsilon = 1$), capital's revenue share *is* γ_m, so firm capital demand is $K_m = \gamma_m\, p_m Y_m / \rho_m$. Taking reform/baseline ratios gives

$$\hat{K}_E \;=\; \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E / \hat{\rho}_E.$$

Here ρ_m bundles the return, depreciation, business taxes, tax depreciation, and any investment tax credit. This channel changes γ_m, not those user-cost terms; in the Philippine solve $\hat{\rho}_E \approx 1$, so the result is read through the near-identity

$$\hat{K}_E \;=\; \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E.$$

Measured for the energy industry on an earlier M=8 steady-state solve of this lever: $\hat{\gamma}_E = +12.2\%$, $\hat{p}_E = -42.8\%$, $\hat{Y}_E = +20.9\%$, so $\hat{K}_E = -22.4\%$; the product $\hat{\gamma}_E \hat{p}_E \hat{Y}_E$ matches the result to 0.03%.

The committed golden records aggregates only (energy output +4.4%, aggregate output +0.001% on the re-blessed run); the sectoral price/capital decomposition is re-derived per solve.

Variables. γ_m, industry m's capital exponent (its share of value added); $\rho_m = (r + \delta - \tau_m^b \delta_m^\tau - \tau_m^{\text{inv}}\delta)/(1-\tau_m^b)$, the cost of capital; p_m, Y_m, the industry's price and output; a hat is a reform/baseline ratio; E marks the energy industry.

2.5 Investment incentive [policy]

Intuition. To pull capital *into* energy, lower the sector's cost of capital, for example with an investment tax credit. That makes energy investment cheaper at the prevailing interest rate, so capital reallocates toward the sector. This is distinct from capital intensity: capital intensity changes the factor *mix* in production, while the credit changes the *cost of capital*. They can move energy capital in opposite directions.

Technical. An investment tax credit τ_m^{inv} on the energy industry enters the cost of capital directly, where the capital exponent is *absent*:

$$\rho_m = \frac{r + \delta - \tau_m^b \delta_m^\tau - \tau_m^{\text{inv}}\delta}{1 - \tau_m^b}.$$

A higher τ_m^{inv} lowers ρ_m, so capital demand $K_m = \gamma_m p_m Y_m/\rho_m$ shifts out at the prevailing r. It is a defined lever, not part of the Philippine reference run.

Variables. τ_m^{inv}, the investment tax credit rate; τ_m^b, the business-income tax; δ, economic depreciation; δ_m^τ, tax depreciation; ρ_m, the cost of capital.

2.6 Health [CLEWS → OG]

Intuition. Cleaner air saves lives and, more important for output, makes working-age people healthier and more productive. A 2.7% cut in power-sector $PM_{2.5}$ does not mean 2.7% fewer pollution deaths. Power is only one source of pollution, and mortality falls less than proportionally as air quality improves, so only about 8% of the emissions cut passes through to deaths. The avoided deaths are concentrated among older people, so the long-run output effect is close to zero. The macro gain comes mainly from working-age productivity during the transition.

Technical. The reform/baseline $PM_{2.5}$ emissions change Δem scales the country's Global Burden of Disease attributable deaths by a calibrated dose-response M:

$$\Delta\text{deaths} \;=\; D_{\text{PM}} \times M \times \Delta\text{em} \;=\; 43{,}951 \times 0.082 \times (-0.0265) \;\approx\; -95.$$

Mortality shocks the age-specific rate $\rho(s,t)$ to hit that signed target and then recomputes the population; *morbidity* scales the effective-labour profile $e(t,s,j)$ by a working-age disability shape. Result: steady-state output changes by −0.0003%, but near-term output rises by +0.19%. The $\times M$ term is what separates −95 from the unadjusted −1,165.

Variables. Δem, the reform/baseline $PM_{2.5}$ emissions change; D_{PM}, GBD ambient-$PM_{2.5}$ attributable deaths (43,951, Philippines); $M \approx 0.082$, the dose-response (energy mass-share × concentration-response elasticity); $\rho(s,t)$, mortality by age s and time t; $e(t,s,j)$, effective labour by time, age, and income group j.

2.7 Discount rate [OG → CLEWS]

Intuition. The economy produces an equilibrium return on capital: its own cost of capital. Passing that return to the energy planner lets CLEWS discount future costs at a rate implied by the macroeconomy rather than by an external assumption. A lower return favours technologies with high up-front costs and long operating lives, such as renewables.

Technical. After the reform solves, OG-Core's equilibrium market return r_p (a multi-period mean) is written to CLEWS's planning discount rate. It is emitted *after* the solve and is one-way: it can change the energy build, but the resulting prices feed back to OG-Core only in a later iteration. By itself it does not change the current OG-Core solve.

Variables. r_p, the market portfolio return (OG-Core's equilibrium cost of capital); the CLEWS discount rate it sets.

2.8 Energy demand [OG → CLEWS]

Intuition. A larger or richer economy needs more energy. After OG-Core solves, its activity level scales the demand path that CLEWS must meet. In a single pass this barely changes, so the channel mainly matters once the two models iterate.

Technical. The reform/baseline activity ratio (industry output Y_m or consumption C_i, raised to an elasticity) scales CLEWS's specified annual demand: $\text{demand}^r = (\hat{Y}_m)^{\varepsilon_D}\,\text{demand}^b$. The signal is emitted after the solve and is one-way. In one pass $\hat{Y}_m \approx 1$, so the demand path is almost unchanged by construction.

Variables. Y_m, industry output; C_i, consumption of good i; ε_D, the demand-scaling elasticity; the CLEWS specified annual demand it scales.